

Lec 31 Motivation for GANs

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Lecture	31 / 63 + intro
Duration	27:55
Captions	English (auto-generated)
Words	4,075
Video	nrzPtPnz2Js
Playlist	Fundamentals of Generative AI and LLMs

Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI large language models. We are in the week five and this week is all about the generative adversarial network and we'll be looking the fundamental concepts which will lay a foundation of for the upcoming week. In the upcoming week we will be having a different variants of GAN. The following are the outline of for the week five. First we'll be looking into the limitations of VAE because the last two weeks we have looked into the we have looked into the VAE. So in this particular week we will first start with what are the limitations of VAE and what is the motivation of for a GAN. Then we will look into the fundamental idea of

[00:47] adversarial learning. Then we will explore the GAN architecture. In the GAN architecture we have generator and discriminator. Then we will move towards the GAN objective and the loss function. How exactly the training happens in a GAN and how do the GAN converge and at something called as equilibrium. We will look into this concept. Towards the end of this particular week we will introduce you to the first variant of GAN called as a DCGAN. That is a deep convolutional GAN. And towards the end of this particular week five we will look into the hands-on and in hands-on we

will see two hands-on sessions. One on basic GAN and the second is on a DCGAN. So these are the outlines for a week five. So in this

[01:35] particular session we will look into two concepts. One is what are the limitations of VAE and what is the motivation of for a GAN. And towards the end of this particular session we will look into the fundamental idea of adversarial learning. So let's get started. So what are the limitations of VAE? Try to recall the concept what we have learned in a week three and a week four. Always in a VAE, the first limitation here is we are doing certain assumptions in the architecture. What are two important assumptions that we are doing in a standard VAE? Recall, is it not the first one that is appropriate posterior is always assumed to be Gaussian.

[02:25] Because you are doing this particular assumption, encoder is going to give us the mean and then the log variance of this Gaussian distribution. So, this is the first assumption that we are doing. And the second assumption that is always the prior. Prior is a generally fixed as the normal distribution or the standard distribution, Gaussian distribution. So, here there are two Gaussians that we are assuming. One is the appropriate posterior is Gaussian and prior is Gaussian. So, because of this, definitely training is going to be very easy. But what if the real data?

[03:07] Because of this assumption, training becomes very easier. But the real world data not necessarily to be following the simple distribution, right? What if the data is more complicated in structure, complicated structure? Then probably this assumption will have an impact. This is the first limitation of VAE and that is Gaussian assumptions. And the second limitation we say that it is a blurry generated images. What exactly is this blurry generated? Just recall from the concept, what is that we are going to reconstruct or generate the data?

[03:46] We'll be generating either the numerical data or if you're generating an the then at a pixel-level we will be comparing the you know X dash $\hat{X}$ and X . So, which means in a standard VAE, VAE usually uses an pixel-wise reconstruction loss. Usually it is a mean squared loss. The to to compare the reconstructed data with the original data X . Now, consider this particular scenario. What is the scenario? I'm passing one input to the encoder. What is that we get? We get the latent representation Z_1 . Now you pass the second input to an encoder and you are getting the Z_2 . Now for some reason, if a Z_1 and Z_2 are equal, so what is the reason? We don't know. But for some reason Z_1 and Z_2 are equal. Now

[04:41] when this is a passed to an decoder, so which means a decoder is now receiving the Z , which are more closer. So, receives nearly the same latent representation Z . In that case, what exactly is the problem to an decoder? So, decoder is now not very sure among the two possible outcomes which is the correct outcome. So, whether it is an image of three or whether it is an image eight. So, the model is not the decoder is not very clear. Why? For one image the model has given one Z . During the training,

the for one image the model has seen one Z. For second image the model has seen a different Z.

[05:30] But here we are getting a similar one. When a Z1 of when for both the images when Z are appropriately closer, then the model is getting confused that what would be the possible outcome. Whether it is an image three or whether it's an image eight. So, when the model is not very sure which is the possible outcome among the two, instead of generating it has to ultimately generate one image. So, what it is going to do? It is going to combine or take an average between those two. So, this averaging what it is going to make, it remove the sharp edges, fine details, thus causes an image to be very blurry. Now, say for example, when you pass this three you are getting one Z. When you pass this eight and you have got the Z

[06:24] which is closer. So, now when you are giving, the model is not very sure because it in a training it has a seen a different Z. But now you are getting a similar Z which the possibility of for this Z can be either three or it can be eight. So, model has already learned that during training. So, now when it is not very sure which is the output that it has to reconstruct, then it is going to take an average. So, when it is [snorts] going to combine or take an average, it is going to produce and a blurry image. So, from this the second limitation is VAE sometimes produces a blurry image. This is the second limitation.

[07:09] Now, let's look into the third limitation. The third limitation is a trade-off between reconstruction and KL divergence. So, a VAE loss function has two terms. One is reconstruction and second is KL divergence. So, what exactly the first term says? The first term says that decoder should reconstruct the input accurately, which means $\hat{X}$ should be approximately equivalent to X . This is what a decoder aim is. If it has to produce an output which is closer to X . What is the other meaning? Encoder has to store good information in a Z. That is what encoder tries to store more information about the input in the latent representation Z.

[07:59] That is all when there is a good latent representation Z, then reconstruction is good, right? Now, what is the KL divergence says? The KL divergence says that this term wants the latent distribution. These two should remain as close as possible. Now, if these two should be should remain closer, what is the other meaning? It prevents the encoder from storing too much of information in the Z. So, it prevents the encoder from storing too much information specific input specific information. So, one is saying store maximum information so that reconstruction will be good. The other one is saying don't store too much information because this should be closer to a normal distribution. So, because of this,

[08:56] improving one may reduce the performance of the other. So, this is the third limitation that is a trade-off. Then, we have the fourth limitation which is called an posterior collapse. » [snorts] » What is this posterior collapse which once again

deals with the KL divergence? Look at this particular term in the loss function. In the loss function, we have reconstruction as well as KL divergence. When is that KL divergence value will be very low?

[09:25] When it is only when these two are closer very much closer or identical, then only KL divergence value will be zero. Now, when this posterior distribution should remain closer to prior, if this is your requirement, because KL divergence should become zero. What is one easy way? One easy because this will have the parameters zero and one. The mean will be zero, the variance is one or the standard deviation the value is one. So, if if you want this to remain closer to this, what is the meaning?

[10:04] Encoder for any encoder has to produce as mean and variance. So, for any input, let encoder give mean zero standard deviation one. So, for every possible input that is given to an encoder, encoder always tries to output mean zero variance one or the standard deviation one. Why to do this? Because the ultimate aim is this should remain closer to this. If you put more pressure, ultimately one easiest way of making these two approximately equal, because the second prior parameters are zero and one. So, for any input give zero and one.

[10:44] Therefore, KL divergence becomes zero. Now, if if this is the case, for every input if the model is giving a mean zero variance one, then is it not definitely KL value of is very small, but latent representation, what about the latent representation? It becomes very uninformative. It is not at all carrying any proper information, because for every input you are giving the same Z. So, rather than same Z, the mean zero and then variance one. So, this is the posterior collapse. What exactly is the problem with the posterior collapse?

[11:25] Look at in a normal VAE, in an ideal case, if the normal VAE, how is that it performs? It takes the input to Z and it gives to an encoder. Encoder produces the mean and then the log variance. From there, Z is sampled using a reparameterization trick and then that Z is then passed to an a decoder and a decoder is going to reconstruct the output. So, which means Z decoder will reconstruct only when a decoder receives a Z and Z is one which has to carry a sufficient information about the input. Then only decoder uses that Z to reconstruct. This is fine.

[12:07] This is an an ideal case. But during a posterior collapse, what is our requirement? Our requirement is that the appropriate posterior distribution should be closer to the prior, which is normal distribution for every input. If that is the case, for input X, it is giving Z1, which is a closer to normal distribution. For input X2, it is giving a Z2, that is closer to a normal distribution. So, which means Z no longer tell the decoder what is that the specific input that Z is carrying.

[12:45] So, there is no proper Z. When there is no proper Z, then how is that the decoder can reconstruct? Right? If Z does not carry the information about X, how is that the decoder will reconstruct? So, what is the solution to this? Whenever there is

an posterior collapse, the solution is decoder is now giving us the reconstructed one irrespective of whatever the Z is passed to an decoder. So, which means decoder is a now reconstructing or generating the data which is independent of Z .

[13:24] Which means then the decoder should be an auto-regressive decoder. In In case of a text, the decoder will be an auto-regressive. In case if it is an image, it is CNN-based non-auto-regressive decoder. So, based on what it has already learned, from that it is going to produce some output, which it definitely does not depends on a Z . This is called posterior collapse. In a short In one line, if you have to explain what is posterior collapse, for every input, this value will be appropriate posterior is closer to prior. Because of this, Z does not carry sufficient information. When Z does not carry sufficient information, then how decoder is going to reconstruct? Decoder is going to reconstruct based on

[14:13] irrespective of whatever Z it is. What it has already learned, based on what it has learned, it is going to produce some output. This is the limitation of fourth limitation of VAE. Then, the fifth limitation. Recall from our week four concept, what is that we have understood? In a VAE, decoder output What is that a decoder will give us? Decoder is going to output the parameters of probability distribution. This This distribution decoder is going to give us the output parameters of this.

[14:50] Now, if it is a continuous data, decoder is going to give us the output, that is $\hat{X}$, and $\hat{X}$ is Here, it is going to give us the mean, and that mean is actually a Gaussian mean of a Gaussian distribution. Suppose, if it is a binary data, then what is that a decoder is going to give us the output? It is going to give us the probability of each binary value. And this is of Bernoulli distribution. If this point is not clear, go back and then recall from our unit from our week three towards the end we have solved one problem and for that problem we had to taken and continuous data and the towards the end of decoder decoder is giving us $\hat{X}$ and how many such values so four such values so four or five such values

[15:41] we had obtained. So those four or five values we actually say it is the mean value and that mean because it is a Gaussian. It is defined as a Gaussian distribution. So now this predefined choices is called an likelihood assumption and this likelihood the choice actually determines the reconstruction loss. So what is the reconstruction loss that we'll be using? Suppose if it is an for continuous data we'll be using mean squared error and for an binary we'll be using binary cross entropy. Okay, so what exactly is an issue? All that is fine, right? So what if something goes wrong with this assumption because if the real world data if it is having more complicated characteristics in the data or if the

[16:30] selected distribution is very smooth, then decoder is not going to capture or going to produce an accurate $\hat{X}$, right? So if the selected likelihood is too simple or does not match the characteristics of the data, then the decoder cannot accurately

represent the output uncertainties accurately. This actually ultimately results in a blurry image or it is a less realistic image. This is the fourth fifth assumption of VAE. Then we have the last limitation and that last limitation is whatever the output that the VAE is going to generate, no doubt it is going to capture to overall structure of the data. But the fine details, um the minute details such as the sharp edges, fine texture, then a small facial details,

[17:28] VAE is not going to capture all of those, you know, details. Therefore, no doubt the sample looks very smooth, but it looks a little less realistic. So, these are certain limitations. And if you look into these limitations, primarily the limitations are due to because you have used you have made certain assumption initially Gaussian assumption and then just sometime back the fifth assumption that is your likelihood assumptions. Then the reconstruction loss because of a reconstruction loss.

[18:04] So, all these leads to an, you know, blurry or unrealistic images. So, with this, what is the motivation of GAN? With all these, if you want to have an another model, then what should be that model? So, ultimately here the limitation is to generate an high realistic data. That is what even VAE requirement also. It has to generate an data which is as close as the original X. So, here the aim is to generate high realistic data, but without explicitly defining the probability distribution. We don't want any probability part to come here.

[18:47] Without explicitly defining the probability distribution or reconstruction loss. Without these two, can I able to generate an image which is more realistic? Then we have to go for a model which is completely without explicitly defining this probability distribution. So, the main motivation of GAN is to generate high realistic data without explicitly explicitly defining a probability distribution or reconstruction loss.

[19:20] So, this is what are the certain limitations of VAE and we have seen what is the motivation for GAN. So, in a GAN, if you look into, it is a generative adversarial learning, GAN. So, in that, let us understand the second term called adversarial learning. What exactly is the main idea behind this and then let us connect it to GAN. So, whenever we say adversarial, is it not the English meaning here is in conflict or competing with each other?

[19:57] So, the meaning of adversarial here is competing with each other. Now, whenever you are competing, definitely you require two. So, the next point here is if at all, if you want to compete with each other, we require two. What are those two? Here we are talking about neural network, so we require two neural networks. Only when there are two neural networks, there can be a competition between that. Therefore, GAN is something where there are two networks. Okay? So, we have now slowly connected to GAN. Now, in GAN, if there are two networks, these two networks learn by competing with each other.

[20:43] Okay. Now, in a GAN, there is G. G stands for generator. Therefore, what is

the GAN? GAN is a neural network which is used to generate the data. So, now, if it has to generate the data, there are there has to be two networks. What are those two networks? The first network we have a generator. We call it as a G. What is the aim of this generator? It is there in the word itself. Generator. The generator has to generate the images. What images it has to generate? It has to generate the fake images. How is that it generates the fake images? From where it generates the fake images? From some random noise. So, generator generates the fake samples or fake data, fake images from the random noise. So, random noise is denoted as a

[21:37] Z. Z is a passed to an generator and generator is going to generate the data and that data is a denoted as $\hat{X}$. Okay. What is the ultimate aim of a generator? It has to generate the data such that it has to look more realistic. Okay. So, generate the samples that look realistic. This is what is the role of a generator. Learn I mean, understand independently later we'll combine it. So, generator is the one which generates the data. What data it has to generate?

[22:10] The fake data. From where it has to generate? It has to generate from random noise. Now, what is the ultimate aim? It has to generate going forward it has to generate a such that it should look realistic. Okay. That's the goal of a generator. Second, GAN we have seen adversarial because there has to be two. Second network what we have here is a discriminator. Now, what is that discriminator? Discriminator is the one which will receive real images as well as a generated images. By receiving real images and generated images, the role of a discriminator is to predict which is real and which is fake. So, we have an discriminator. To that we are giving an input X which can be real or a fake and the discriminator has to tell which is a

[23:03] real, which is a fake. So, what is the goal of a discriminator here? The goal of a discriminator is Correctly, it has to tell which is a real image and which is a generated image. It has to correctly distinguish real samples from generated samples. Now, if you're looking into Is it not there is uh there are two networks here. One network is generator, second network is discriminator. Generator has to generate and it has to generate a fake images. Um and the the role of a generator is or the aim of a generator here is it has to generate the image as realistic as possible. What is the role of a discriminator? Discriminator has to receive both real images as well as fake images. And ultimately, discriminator

[23:51] has to tell which is real and which is a fake. So, now generator will put more efforts to generate realistic image so the discriminator should fail to identify. And discriminator becomes a stronger and stronger in order to identify which is real and which is a fake. So, because of this, are you now seeing there is some sort of a competition between these two? Okay. Now, let us try to understand how the competition works here. So, first what is that we have? We have an discriminator. What is the role of a discriminator? It has to learn to distinguish real samples from a fake samples which are generated by the generator. See, discriminator has to learn

to distinguish real samples from fake samples. And who generates these

[24:39] fake samples? By the generator. At the other side, we have generator. So, what is the role of a generator? Generator has to learn to generate more realistic samples so that discriminator cannot easily identify which is a generated image or which is a real image. So, this is the two networks, discriminator and generator. What is a learning process here? So, the learning process is a first generator has to generate the fake images and discriminator has to classify it as a real or a fake. Then discriminator, whatever discriminator has a predicted, discrim- uh generator will take that as a feedback and generator will improve.

[25:26] So, here the learning process is generator generates a fake samples, discriminator classify it as a real or fake, and the generator uses a discriminator's prediction to improve. And this process is repeated. So, through this process, over a period of time, is it not generator gradually learns to uh generate the images which are more realistic? Because just say, if discriminator can able to tell it is a fake image or it is a generated image, and if that feedback is given to an generator, then generator can able to take that feedback and it can able to improve by itself.

[26:10] The over a period of time, gradually, is it not that um you know, the generator gradually learns to generate more realistic samples while discriminator becomes better at identifying whether it is real or a fake? So, is it not an adversarial learning here? Adversarial is two networks competing with each other. One is a generator, second is a discriminator. The role of a generator is to generate the samples more realistically. Gener- discriminator has to classify which is real and which is generated one.

[26:47] » [snorts] » So, uh generator constantly put the efforts to improve again and again so that it can generate more realistically. Discriminator will put more efforts to identify which is a real or which is a fake. So, with this both will be able to at the end we are able to get the image which is more realistic to the original image. So, this is all about the adversarial learning. So, with this we have come to an end of this session where in this session we have primarily focused only on what are the limitations of VAE and what is the motivation of a generator.

[27:28] And we have a seen an introductory session of an adversarial learning. So, in the next session we will look into a detailed architecture of GAN where we will see how generator works, how discriminator works, and then we will also see what is a loss function and then the back propagation of GAN. » » Thank you.